

1.INTRODUCTION TO EMBEDDED SYSTEMS

What is an Embedded System?

An embedded system is a combination of computer hardware and software designed to perform a specific, dedicated function within a larger mechanical or electrical system. Unlike general-purpose computers (like laptops or smartphones), embedded systems are built for a single purpose and are optimized for size, cost, and power consumption.

They typically consist of a microcontroller/microprocessor, memory, input sensors, and output actuators, all working together with embedded software (firmware) to perform real-time operations.

REAL-WORLD APPLICATIONS

1.SMART HOME: Smart thermostats, smart lighting, smart door locks, and home security systems.

2.AUTOMOTIVE: Engine control units (ECUs), anti-lock braking systems (ABS), airbag control, and infotainment systems.

MICROCONTROLLER VS MICROPROCESSOR

Feature	Microcontroller	Microprocessor
Definition	A complete computing system on a single chip (CPU + RAM + ROM + I/O)	Only the CPU; needs external RAM, ROM, and I/O components
Cost	Low cost, ideal for dedicated tasks	Higher cost, used for general-purpose computing
Power Consumption	Low power	Higher power
Example	Arduino UNO, ESP32, Raspberry Pi Pico	Intel i5/i7, ARM Cortex-A series (used in PCs/smartphones)
Use Case	Embedded, single-function devices	Multi-tasking systems like computers

2.COMPONENT STUDY

Role of Microcontrollers:

- 1. Arduino UNO: An 8-bit AVR-based microcontroller board, widely used for beginner embedded projects. It has digital/analog I/O pins for reading sensors and controlling actuators like LEDs, motors, and buzzers.**
- 2. ESP32: A more powerful microcontroller with built-in Wi-Fi and Bluetooth, making it ideal for IoT applications where data needs to be sent to the cloud or a mobile app.**
- 3. Raspberry Pi Pico: A low-cost microcontroller board based on the RP2040 chip, offering dual-core processing, suitable for slightly more complex embedded tasks while remaining power-efficient.**

BASIC SENSORS

- 1. TEMPERATURE SENSOR:** Measures ambient temperature and converts it into an electrical signal that the microcontroller can read and process.
- 2. MOTION SENSOR :** Detects infrared radiation changes caused by movement of humans/objects, commonly used in security and automation systems.
- 3. Light Sensor:** Detects ambient light intensity, used in automatic street lighting and smart lighting systems.

Simulation code :

```
#include <DHT.h>

#define DHTPIN 2

#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

#define GREEN_LED 8
#define RED_LED 9
#define BUZZER 10

float threshold = 30.0; // Temperature threshold in Celsius

void setup() {
  Serial.begin(9600);
  dht.begin();

  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);
}

void loop() {
  float temperature = dht.readTemperature();

  if (isnan(temperature)) {
    Serial.println("Failed to read from DHT sensor!");
    delay(2000);
    return;
  }
}
```

```
Serial.print("Temperature: ");
```

```
Serial.print(temperature);
```

```
Serial.println(" °C");
```

```
if (temperature > threshold) {
```

```
    digitalWrite(RED_LED, HIGH);
```

```
    digitalWrite(GREEN_LED, LOW);
```

```
    digitalWrite(BUZZER, HIGH);
```

```
    Serial.println("ALERT: Temperature threshold exceeded!");
```

```
} else {
```

```
    digitalWrite(RED_LED, LOW);
```

```
    digitalWrite(GREEN_LED, HIGH);
```

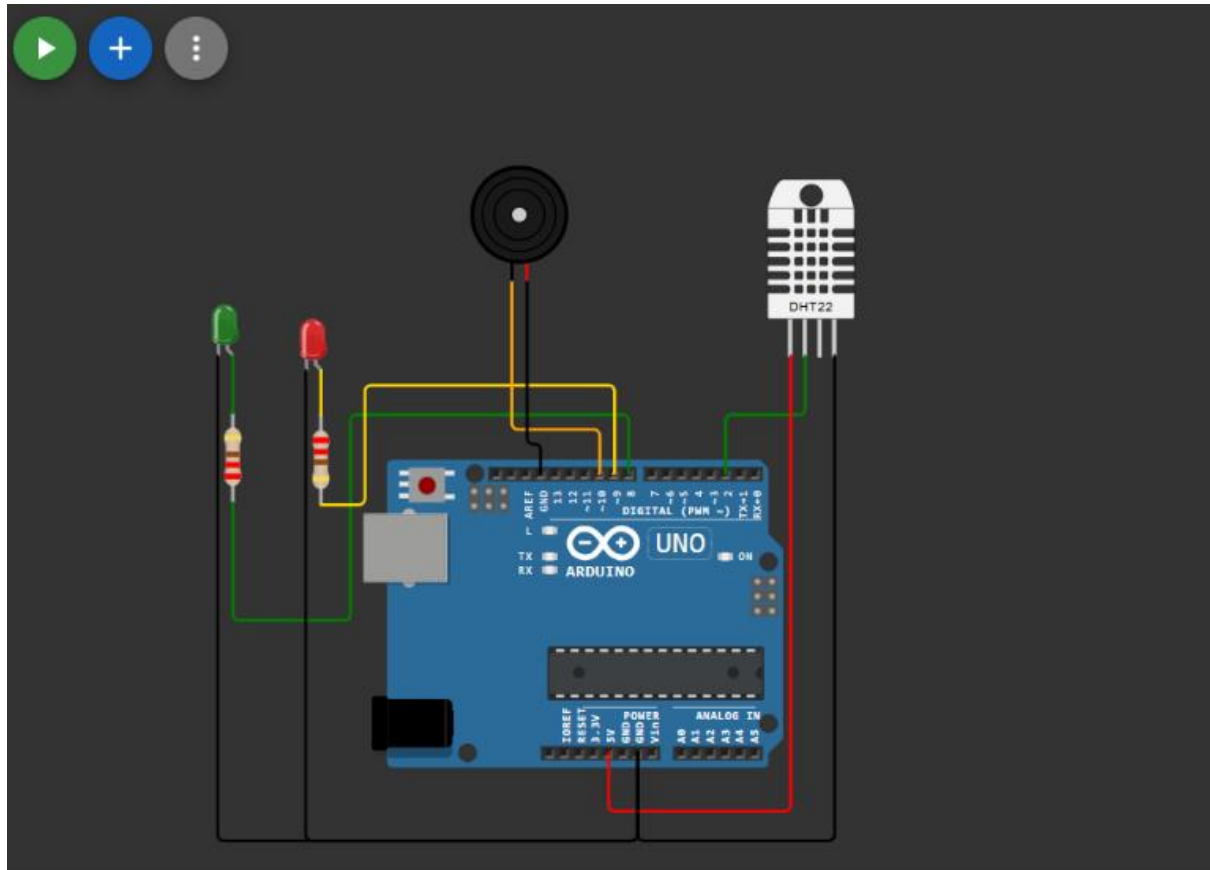
```
    digitalWrite(BUZZER, LOW);
```

```
}
```

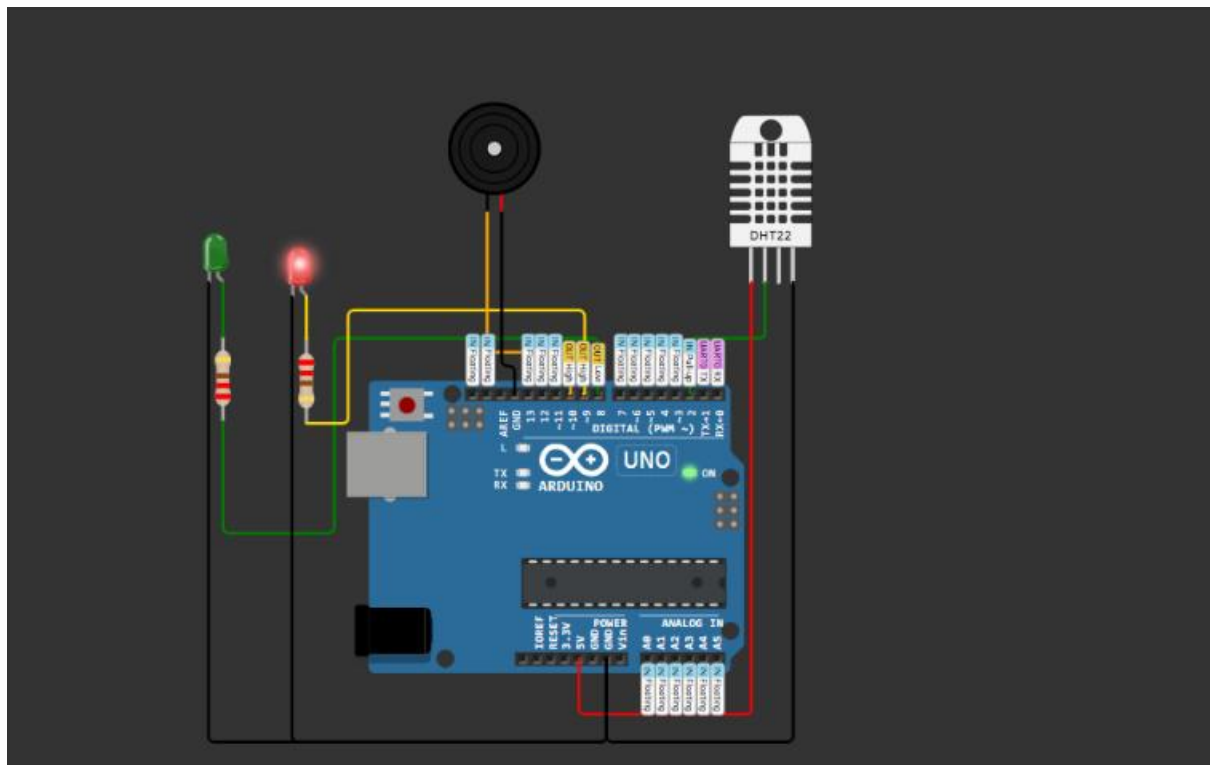
```
delay(2000);
```

```
}
```

CIRCUIT DIAGRAM

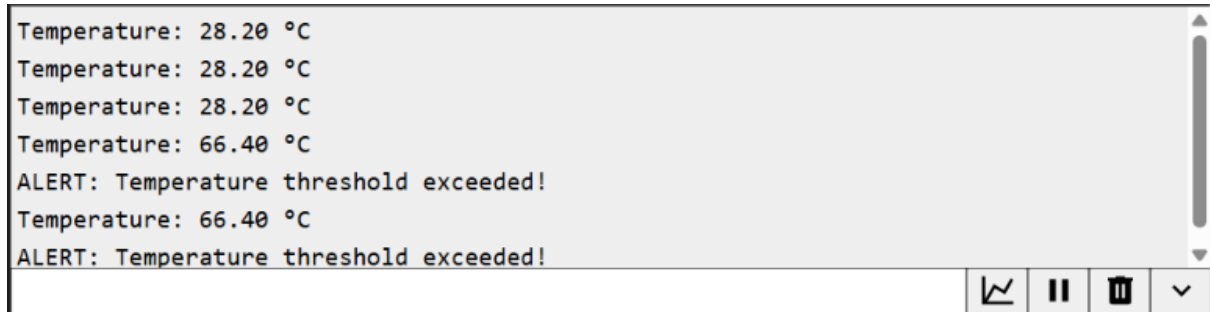


SIMULATION



OUTPUT

```
Temperature: 28.20 °C
Temperature: 28.20 °C
Temperature: 28.20 °C
Temperature: 66.40 °C
ALERT: Temperature threshold exceeded!
Temperature: 66.40 °C
ALERT: Temperature threshold exceeded!
```

A terminal window with a light gray background and a dark border. It contains seven lines of text: three 'Temperature: 28.20 °C' lines, one 'Temperature: 66.40 °C' line, one 'ALERT: Temperature threshold exceeded!' line, another 'Temperature: 66.40 °C' line, and another 'ALERT: Temperature threshold exceeded!' line. On the right side of the terminal, there is a vertical scrollbar. At the bottom right of the terminal, there is a toolbar with four icons: a line graph, a pause symbol, a trash can, and a dropdown arrow.

			
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